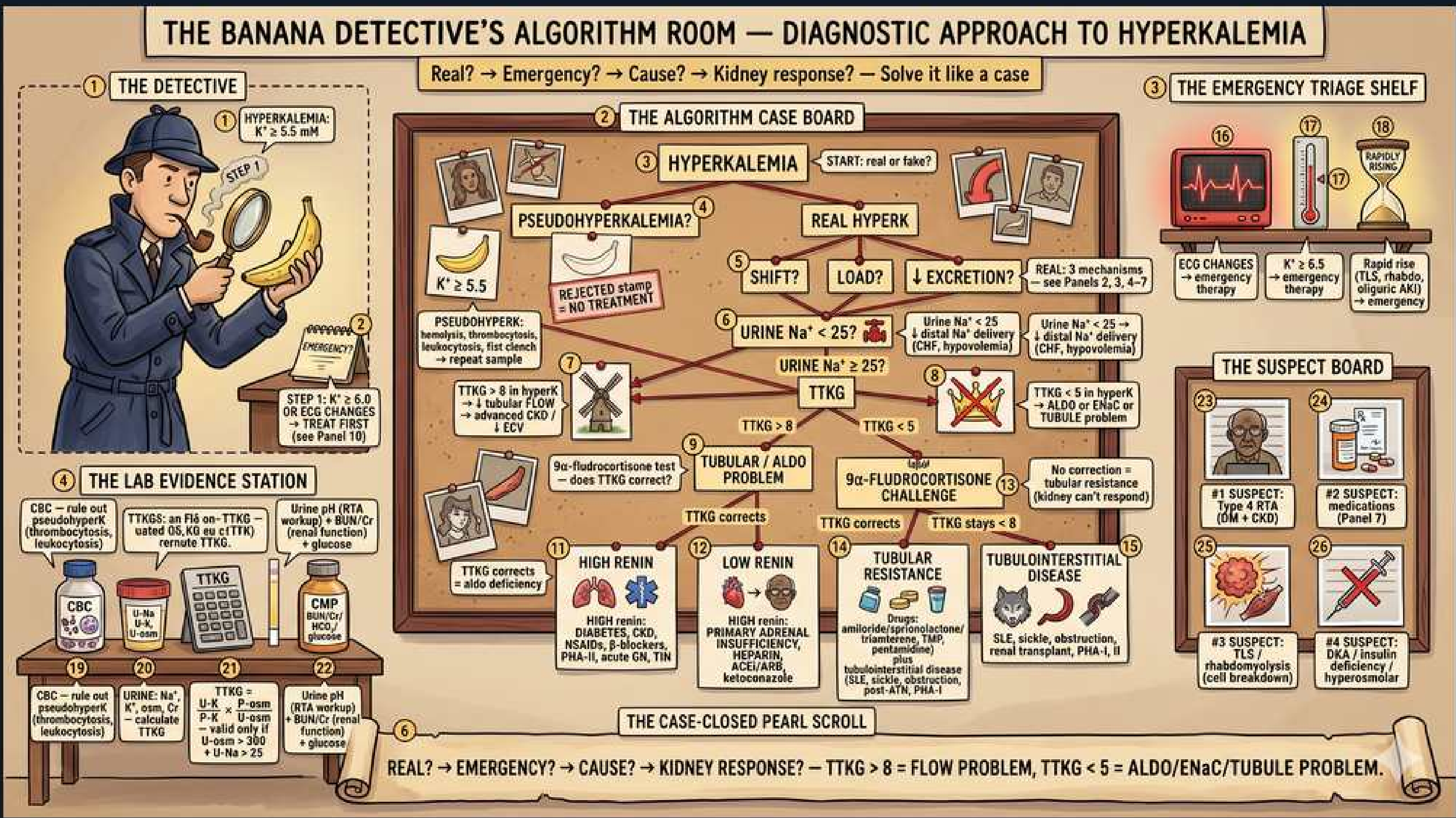
Hyperkalemia: Diagnostic Algorithm & TTKG Approach



1. Definition K+ >5.5

WHAT YOU SEE

Detective with banana, 'HYPERKALEMIA K+ >5.5'

WHAT IT ENCODES

Hyperkalemia is serum K+ >5.5 mEq/L. Severity guides urgency: mild 5.5-6.0, moderate 6.1-7.0, severe >7.0 or any ECG changes. Approach: is it real → emergency → cause → kidney response.

BOARD TRAP

A single high value without ECG changes may still be life-threatening if rising fast — rate of change matters, not just the number.

2. Pseudohyperkalemia

WHAT YOU SEE

'PSEUDOHYPERKALEMIA?' rejected portrait

WHAT IT ENCODES

Rule out pseudohyperkalemia first: hemolyzed sample, fist clenching/tourniquet, marked thrombocytosis or leukocytosis releasing K+ in the tube. If suspected, redraw — no treatment needed.

BOARD TRAP

Thrombocytosis/leukocytosis elevates SERUM (clotted) K+; a normal PLASMA (heparinized) potassium confirms pseudohyperkalemia.

3. ECG changes = emergency

WHAT YOU SEE

Red ECG/EKG icon, 'EMERGENCY THERAPY'

WHAT IT ENCODES

ECG progression: peaked T waves → PR prolongation → flattened/absent P → widened QRS → sine wave → arrest. Any ECG change mandates immediate IV calcium to stabilize the myocardium.

BOARD TRAP

Calcium does NOT lower potassium — it stabilizes cardiac membranes; you still need shift + removal therapies.

4. Calcium membrane stabilization

WHAT YOU SEE

Lightning/ECG shelf, emergency therapy

WHAT IT ENCODES

IV calcium gluconate/chloride is first for ECG changes — onset in minutes, lasts ~30-60 min, raises the threshold potential to prevent arrhythmia. Repeat if ECG doesn't improve.

BOARD TRAP

Use calcium cautiously in digoxin toxicity (risk of 'stone heart') — give slowly/infusion; it is not absolutely contraindicated.

5. Shift therapy (insulin)

WHAT YOU SEE

'SHIFT?' algorithm node

WHAT IT ENCODES

Shift K+ intracellularly: insulin (with glucose) is most reliable, plus inhaled beta-agonist (albuterol) and, in acidosis, sodium bicarbonate. Onset minutes; effect transient — does not remove potassium.

BOARD TRAP

Insulin shifts but doesn't remove K+ — always pair with a removal strategy; monitor for hypoglycemia after insulin.

6. Increased load

WHAT YOU SEE

'LOAD?' node

WHAT IT ENCODES

Increased K+ load: rhabdomyolysis, tumor lysis, hemolysis, GI bleed, massive transfusion, exogenous intake. Normal kidneys excrete an acute load; load-related hyperkalemia usually implies impaired excretion too.

BOARD TRAP

In tumor lysis, hyperkalemia accompanies hyperphosphatemia, hyperuricemia, and HYPOcalcemia — treat the full electrolyte panel.

7. Decreased excretion

WHAT YOU SEE

'↓EXCRETION?' node, '3 mechanisms'

WHAT IT ENCODES

Reduced renal excretion (most chronic cause): low GFR (AKI/CKD), low aldosterone effect, or reduced distal Na+ delivery. The TTKG then localizes the defect.

BOARD TRAP

Distinguish low aldosterone (hyporeninemic hypoaldosteronism, type 4 RTA) from aldosterone RESISTANCE — TTKG and renin/aldo levels separate them.

8. Urine K+ >25 (renal intact)

WHAT YOU SEE

'Urine Na+ ≥25?' / urine K+ branch

WHAT IT ENCODES

Appropriate renal response to hyperkalemia is kaliuresis (urine K+ high, TTKG >7). A high TTKG points to a FLOW/load problem rather than a tubular secretory defect.

BOARD TRAP

Adequate urinary K+ excretion shifts the differential toward intake/cell-shift causes, away from a tubular defect.

9. TTKG calculation

WHAT YOU SEE

'TTKG' branch nodes in algorithm

WHAT IT ENCODES

Transtubular K gradient = (urine K × plasma osm)/(plasma K × urine osm). TTKG >7 = appropriate aldosterone-driven secretion; TTKG <5 in hyperkalemia = impaired secretion (aldosterone or tubule problem).

BOARD TRAP

TTKG is only valid when urine osm > plasma osm and urine Na >25 — otherwise the gradient is uninterpretable.

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10. TTKG >7 = flow problem

WHAT YOU SEE

'TTKG >7 → flow problem' left branch

WHAT IT ENCODES

High TTKG with hyperkalemia means secretion works but flow/delivery or load is the issue — e.g., ACEi/ARB, volume depletion limiting distal Na delivery. Aldosterone-tubule axis is intact.

BOARD TRAP

A high TTKG essentially excludes a primary tubular secretory defect — the lesion is upstream (flow/RAAS/intake).

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11. TTKG <5 = tubule/aldo problem

WHAT YOU SEE

'TTKG <5 → ALDO or tubule problem' branch

WHAT IT ENCODES

Low TTKG indicates inadequate secretion: either aldosterone deficiency or ENaC/tubule resistance. Next step: fludrocortisone challenge — correction implies deficiency, non-correction implies resistance.

BOARD TRAP

Fludrocortisone response is the discriminator: corrects = hypoaldosteronism; fails = tubular resistance (PHA-1) or drug block (amiloride).

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12. High renin (Addison/CAH)

WHAT YOU SEE

'HIGH RENIN' branch, 'DIABETES, CKD, Addisons'

WHAT IT ENCODES

High-renin hyperkalemia: primary adrenal failure (Addison), CAH (21-hydroxylase deficiency), or ACEi/ARB blocking angiotensin. Aldosterone is low despite high renin drive.

BOARD TRAP

Addison's gives hyperkalemia WITH hyponatremia and hypotension — distinguish from isolated type-4 RTA (normo/hypertensive, normal Na).

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13. Low renin (type 4 RTA)

WHAT YOU SEE

'LOW RENIN' branch, 'DIABETES, NSAIDs, ...'

WHAT IT ENCODES

Hyporeninemic hypoaldosteronism (type 4 RTA): diabetic nephropathy, NSAIDs, calcineurin inhibitors. Low renin → low aldosterone → mild hyperkalemia with normal anion gap metabolic acidosis.

BOARD TRAP

Type 4 RTA = HYPERkalemic normal-gap acidosis — contrast with type 1/2 RTA which are HYPOkalemic; potassium direction is the key separator.

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14. Tubular resistance / drugs

WHAT YOU SEE

'TUBULAR RESISTANCE' branch, spironolactone

WHAT IT ENCODES

Aldosterone resistance: K+-sparing diuretics (spironolactone, eplerenone, amiloride, triamterene), trimethoprim, pentamidine block ENaC/receptor. Fludrocortisone fails to correct.

BOARD TRAP

Spironolactone + ACEi/ARB is a common iatrogenic hyperkalemia combination — review the med list before chasing rare causes.

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15. Case-closed pearl: □5 rule

WHAT YOU SEE

Bottom scroll 'TTKG >5 flow problem, <5 aldo/tubule'

WHAT IT ENCODES

Summary algorithm: Real? → Emergency (ECG/calcium)? → Cause? → Kidney response via TTKG: >7-8 = flow/load problem; <5 = aldosterone or tubule problem. This sequence solves any hyperkalemia case.

BOARD TRAP

Skipping the 'is it real' and 'is it an emergency' steps to chase etiology can miss pseudohyperkalemia or a lethal arrhythmia.
